Project Grading Criteria & Specification

Grading Checklist

Each criterion is graded on a scale of **0-4 points**. A justification is required for every score assigned.

1. Problem Formulation [0-4 pts]

- **Problem Statement [1 pt]:** Is the problem clearly stated?
- **Objectives/Use Cases [1 pt]:** Is the point of creating the model clear, and are potential use cases defined?
- **Data Source/Description [1 pt]:** Where does the data come from, and what does it contain?
- **Preprocessing [1 pt]:** Is the preprocessing step clearly described?

2. Model [0-4 pts]

- **Two Models [1 pt]:** Are two different models specified?
- **Explanation of Differences [1 pt]:** Are the differences between the two models explained?
- **Justification [1 pt]:** Is the difference in the models justified (e.g., does adding an additional parameter make sense)?
- **Technical Description [1 pt]:** Are models sufficiently described (formulas, parameters, required data)?

3. Priors [0-4 pts]

- **Prior Selection Rationale [1 pt]:** Is it explained why particular priors for parameters were selected?
- **Prior Predictive Checks (Parameters) [1 pt]:** Have prior predictive checks been done for parameters (do simulated parameters make sense)?
- **Prior Predictive Checks (Measurements) [1 pt]:** Have prior predictive checks been done for measurements (do simulated measurements make sense)?
- **Selection Method [1 pt]:** How were prior parameters selected?

4. Posterior Analysis (Model 1) [0-4 pts]

- **Sampling Issues [1 pt]:** Were there any issues with sampling? If so, what mitigation ideas were used?
- **Posterior Predictive Distribution [1 pt]:** Are the samples from the posterior predictive distribution analyzed?
- **Data Consistency [1 pt]:** Are data consistent with posterior predictive samples, and is it sufficiently commented (or justified if inconsistent)?
- **Marginal Distributions [1 pt]:** Have parameter marginal distributions been analyzed (histograms, summaries, concentration/diffusion analysis)?

5. Posterior Analysis (Model 2) [0-4 pts]

- **Sampling Issues [1 pt]:** Were there any issues with sampling? If so, what mitigation ideas were used?
- **Posterior Predictive Distribution [1 pt]:** Are the samples from the posterior predictive distribution analyzed?
- **Data Consistency [1 pt]:** Are data consistent with posterior predictive samples, and is it sufficiently commented (or justified if inconsistent)?
- **Marginal Distributions [1 pt]:** Have parameter marginal distributions been analyzed (histograms, summaries, concentration/diffusion analysis)?

6. Model Comparison [0-4 pts]

- **Information Criteria Usage [1 pt]:** Have models been compared using information criteria?
- **WAIC Discussion [1 pt]:** Have results for WAIC been discussed (winner, overlap, warnings)?
- **PSIS-LOO Discussion [1 pt]:** Have results for PSIS-LOO been discussed (winner, overlap, warnings)?
- **Final Assessment [1 pt]:** Was the comparison discussed? Do the authors agree with the criteria? What is the rationale for selecting one model over the other?

Project Specification

*Note: The report structure does not have to strictly match the workflow (e.g., a second model might be specified after identifying deficiencies in the first). Grading focuses on the **thought process**, not just benchmarks. Models performing poorly are acceptable if there were sound reasons to test them.*

Requirements

1. **Team Formation:** Form 2-person teams.
2. **Phenomena Selection:** Select a phenomenon to model and obtain relevant data.
3. **Modeling:** Propose two models to represent the phenomenon.
4. **Prior Analysis:** Perform prior analysis and selection.
5. **Fitting & Analysis:** Fit models and analyze posterior predictive distributions (check if they capture the data). Repeat steps 3-5 if necessary.
6. **Comparison:** Perform a comparison of models using WAIC and LOO information criteria.
7. **Report:** Provide a written report (generated from Jupyter Notebooks) and prepare a presentation for the final classes.

IMPORTANT: each topic has to be accepted by your assigned teacher.

Grading Weighting

- **Team 1 Assessment:** 25%
- **Team 2 Assessment:** 25%
- **Teacher Assessment:** 50%